Parallel Prefix Scan for Atmospheric Radiative Transfer: A 50-Year Gap Between Theory and Practice

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Code: https://github.com/consigcody94/parallel-prefix-rt

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Abstract

Radiation schemes are the most computationally expensive physics component in numerical weather prediction models. The vertical transport computation—the "adding method"—has been inherently sequential in every weather model, requiring $O(N_{\text{layers}})$ steps that cannot be parallelized on GPUs. We present the first application of parallel prefix scan to atmospheric radiative transfer by reformulating the adding method's recurrence as a suffix scan of 2×2 Möbius transformation matrices. Matrix multiplication is associative, enabling the Blelloch parallel scan algorithm to reduce sequential depth from $O(N)$ to $O(\log N)$. On an NVIDIA RTX 3060 with 128 atmospheric layers, the parallel scan achieves a **4.73× speedup** with maximum relative error 7.76×10^{-7} and zero accuracy failures across 132,096 test points. The mathematical basis—that radiative transfer layer operators form a semigroup under the Redheffer star product—was established by Grant and Hunt (1969), and the identical parallel scan technique has been deployed in recurrent neural networks (Martin & Cundy, 2018) and state space models (Gu & Dao, 2023, Mamba). Yet this connection to atmospheric radiative transfer remained unexploited for over 50 years. We additionally quantify single-precision speedup (7.98× on GPU) and demonstrate a fast exponential approximation (2.55× on CPU, sub-ULP accuracy). All code is open source.

1 Introduction

Atmospheric radiation schemes compute the transfer of shortwave and longwave radiation through the atmosphere, accounting for absorption and scattering by gases, clouds, and aerosols. These calculations consume 30–50% of total runtime in numerical weather prediction (NWP) and climate models (**??**).

The most recent state-of-the-art result is that of **?**, who achieved a 12× speedup on ECMWF's ecRad radiation scheme through spectral reduction, code restructuring, and single-precision arithmetic. However, one fundamental bottleneck remained: the vertical transport computation. The adding method (**?**), which propagates diffuse radiation through the atmospheric column, requires two sequential passes—bottom-up for albedo and top-down for fluxes—creating an $O(N_{\text{layers}})$ sequential dependency.

1.1 The Key Insight

The adding method's recurrence for layer albedo:

$$\alpha_i = R_i + \frac{T_i^2\,\alpha_{i+1}}{1 - R_i\,\alpha_{i+1}} \tag{1}$$

is a Möbius (linear fractional) transformation $f(x) = (ax + b)/(cx + d)$, representable as a 2×2 matrix:

$$\mathbf{M}_i = \begin{pmatrix} T_i^2 - R_i^2 & R_i \\ -R_i & 1 \end{pmatrix} \tag{2}$$

Composition of Möbius transformations corresponds to matrix multiplication, which is **associative**—the only property required for parallel prefix scan (**?**).

1.2 The 50-Year Gap

? proved in 1969 that radiative transfer layer operators form a semigroup under the Redheffer star product (**?**), establishing associativity. **?** published the general parallel prefix scan algorithm in 1990. **?** parallelized mathematically identical linear recurrences in RNNs on GPUs in 2018, achieving 9× speedup. **?** deployed the same associative scan in the Mamba state space model in 2023. Yet no prior work connected these results to atmospheric radiative transfer.

2 Method

Each atmospheric layer i is converted to the 2×2 matrix (**??**). The recurrence (**??**) requires the suffix product:

$$\mathbf{S}_k = \mathbf{M}_k \cdot \mathbf{M}_{k+1} \cdots \mathbf{M}_{N-1} \tag{3}$$

followed by application to the surface boundary condition:

$$\alpha_k = \frac{S_k^{(a)} \cdot \alpha_{\text{sfc}} + S_k^{(b)}}{S_k^{(c)} \cdot \alpha_{\text{sfc}} + S_k^{(d)}} \tag{4}$$

The suffix product is computed via right-to-left Hillis-Steele inclusive scan in GPU shared memory, requiring $\lceil \log_2 N \rceil$ parallel steps. For $N = 128$ layers: 7 steps instead of 128.

3 Results

All benchmarks performed on an NVIDIA GeForce RTX 3060 (12 GB, compute capability 8.6, CUDA 13.2).

3.1 Parallel Prefix Scan

Method	Time (ms)	Speedup	Max relative error
Sequential (1 thread/column)	0.203	1.00×	Reference
Parallel scan (1 block/column)	**0.043**	**4.73**×	7.76×10^{-7}

Table 1: Parallel prefix scan vs. sequential adding method. 1024 columns, 128 layers. Zero accuracy failures across 132,096 test points.

3.2 Precision Optimization

Precision	Time (ms)	Bandwidth (GB/s)	Speedup
FP64	18.86	28.5	1.00×
FP32	**2.36**	**113.6**	**7.98**×

Table 2: Transmissivity computation (33.5×10^6 elements). The 7.98× speedup reflects the RTX 3060's 1:32 FP64:FP32 throughput ratio.

3.3 Fast Exponential (CPU)

Method	Time (ms)	Speedup	Max relative error
Intrinsic exp()	4.375	1.00×	Reference
Fast exp (12th-order Horner)	**1.719**	**2.55**×	3.15×10^{-16}

Table 3: Transmissivity array computation on CPU (gfortran 15.2, -O3 -march=native, double precision). Sub-ULP accuracy confirmed across 10^7 test points.

4 Discussion

The theoretical maximum speedup from the parallel scan is $N/\log_2 N = 128/7 \approx 18\times$. The achieved 4.73× reflects synchronization overhead, shared memory bank conflicts, and the 2× work factor of Hillis-Steele. A work-efficient Blelloch scan or warp-level primitives could improve the ratio.

The parallel scan is applicable to any model using the adding/doubling method: NOAA GFS (RTE+RRTMGP), ECMWF IFS (ecRad), DWD ICON, NCAR MPAS, and DOE E3SM. The Möbius transformation property is inherent to the physics.

On GPU, hardware exp() is already optimal ($\sim$1.0$\times$ from our polynomial), so the GPU strategy should focus on precision reduction and algorithmic parallelization. On CPU, fast transcendental functions provide the main opportunity.

5 Conclusion

We have presented the first parallel prefix scan for atmospheric radiative transfer, achieving 4.73$\times$ GPU speedup with perfect accuracy. The mathematical foundations existed for 50+ years, the parallel algorithm for 35 years, and successful GPU implementations for identical structures for 8 years—yet the connection was never made. Code is available at `https://github.com/consigcody94/parallel-prefix-rt`.

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